CAFÉ SHOP PERFORMANCE

Recommended Analysis & KPI Framework

This document defines the business questions and key performance indicators (KPIs) to guide the café data analysis project. It is designed as a starting point for exploring the dataset, building Excel PivotTables and measures, creating an interactive dashboard, and communicating findings to business stakeholders.

PROJECT OBJECTIVE
Understand sales performance, product mix, demand patterns, and store-level differences to identify practical opportunities for better staffing, inventory planning, and sales optimisation.

1. Recommended Business Questions

Analysis area	Recommended question
Sales performance	How much sales revenue is generated, and how does it vary by month, weekday, product category, and store?
Hourly demand	Which hours have the highest and lowest quantity or transaction volumes? Are staffing and preparation aligned with demand?
Product performance	Which products and categories contribute most to sales? Which products may need a closer review of demand and profitability?
Order-size mix	How are orders distributed across size groups? Is the 'Not Defined' segment small enough for reliable analysis?
Store comparison	Which locations lead in sales and transaction volume? Do differences remain after considering operating hours, labour costs, and margins?
Basket behaviour	What are the average bill and average items per transaction? Could bundles increase basket value without reducing margin?
Data quality	Are transaction IDs, dates, product categories, quantities, and store names complete and consistently recorded?

2. Core KPIs and Definitions

KPI	Definition / calculation	Unit
Total Sales	Sum of the sales/bill amount for all included records.	Currency
Total Quantity Sold	Sum of item quantities sold.	Items
Transaction Count	Count of valid transactions; use distinct transaction IDs if IDs repeat across line items.	Transactions
Footfall	Number of customer visits, only if the source dataset records visits separately from transactions.	Visits
Average Bill	Total sales ÷ valid transaction count. Confirm tax, discounts, and refunds treatment.	Currency / transaction
Average Order Size	Total quantity sold ÷ valid transaction count; represents items per transaction.	Items / transaction
Sales by Category	Total sales grouped by product category.	Currency and % of total
Sales by Product	Total sales grouped by product or item name.	Currency and rank
Hourly Demand	Quantity sold or transaction count grouped by hour of day.	Items / transactions

KPI	Definition / calculation	Unit
Orders by Weekday	Valid transaction count grouped by weekday.	Transactions
Store Performance	Sales and transaction count grouped by store location.	Currency and transactions
Order Size Distribution	Count and percentage of transactions in each size group, including missing/undefined values.	Count and %

3. Recommended Dashboard Views

Dashboard element	Recommended design
KPI cards	Total Sales, Total Quantity, Transaction Count or Footfall (clearly defined), Average Bill, Average Order Size.
Hourly demand line chart	Show quantity or transactions by hour to identify peak and low-demand periods.
Category sales bar chart	Compare sales by product category; optionally show percentage contribution.
Top products chart	Rank leading products by sales and label the measure and time period.
Order-size distribution	Use a bar or doughnut chart to show counts/percentages, including Not Defined.
Weekday orders chart	Compare transaction count by weekday using a consistent axis.
Store comparison	Compare sales and transactions with clearly labelled measures; avoid mixing currency and counts on an unclear axis.
Slicers	Use date/month and weekday slicers where appropriate; ensure all relevant PivotTables respond consistently.

4. Suggested Analysis Workflow

• **Load and inspect:** review the source tables, column names, data types, row count, and date coverage.

• **Clean:** check missing values, duplicate records, invalid quantities, inconsistent product labels, and date/time formats.

• **Validate:** confirm transaction ID uniqueness and determine whether each row is a transaction or a transaction line item.

• **Build KPIs:** calculate total sales, quantity, transaction count, average bill, average order size, and category/store summaries.

• **Explore:** answer the recommended questions using PivotTables, charts, and appropriate calculated measures.

• **Visualise:** create a clear dashboard with consistent labels, meaningful chart types, and interactive slicers.

• **Communicate:** prepare a business report and PowerPoint with evidence-based findings, recommendations, and limitations.

5. Initial Results to Validate

The supplied dashboard screenshots show the following reported values. Treat these as initial reference values and reconcile them against the cleaned source data before publishing the project or presenting it in an interview.

Metric	Reported value
Total Sales	$698,812.33
Total Quantity Sold	214,470
Reported Footfall / Transaction ID Count	149,116
Average Bill	$4.69
Average Order Size	1.44 items

Important validation note: The screenshots use 149,116 for both “Total Footfall” and the count of transaction_id. Confirm whether this is transaction count, customer visits, or another metric. Also verify that Average Bill and Average Order Size use the intended denominator and that charts display the correct units.

6. Expected Deliverables

- Cleaned and analysis-ready Excel dataset
- PivotTables and KPI calculations / measures
- Interactive Excel dashboard
- Business analysis report with findings and recommendations
- PowerPoint presentation summarising the project

Prepared for a portfolio data analytics project. Findings should be described as observations from the dataset; causal explanations and profitability claims require supporting cost, margin, and operational data.